

AMITH C KOWSHIK

College Park, MD

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Education

University of Maryland, College Park

Master of Science in Artificial Intelligence

Expected: May 2028

College Park, MD

Relevant Coursework: Principles of Machine Learning, Probability and Statistics, AI and Society

Bangalore Institute of Technology

Bachelor of Engineering in Computer Science and Engineering

Dec. 2020 – Jun. 2024

Bangalore, India

CGPA: 8.25/10

Relevant Coursework: Data Structures & Applications, Design & Analysis of Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Automata Theory and Computability, System Software & Compiler Design, Artificial Intelligence & Machine Learning, Web Design

Technical Skills

Languages: Golang, C, C++, Java, Python, TypeScript, JavaScript

Databases: MySQL, PostgreSQL, Redis

Frameworks & Libraries: React, Angular, Django, Node.js, Tailwind CSS

Cloud & Infrastructure: Docker, Kubernetes, AWS

Monitoring & Observability: Prometheus, Grafana, Alertmanager

Version Control: Git, GitHub, GitLab

Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV, HuggingFace

Experience

F5 Networks

Jul. 2024 – Aug. 2026

Software Development Engineer II

Bangalore

- Engineered network defense and private cloud products using Golang microservices with REST APIs, enabling distributed deployments across large-scale clusters while incorporating system design principles.
- Delivered a managed DDoS protection platform for enterprise customers, strengthening resilience against large-scale attacks and improving service scalability.
- Processed millions of requests per day with greater than 99.99% uptime, ensuring reliability under heavy traffic.
- Implemented internal monitoring and observability tools, accelerating incident detection by approximately 30%.
- Automated monitoring pipelines, reducing false positives by approximately 20% and enhancing visibility across engineering teams.

F5 Networks

Feb. 2024 – May 2024

Technical Intern

Bangalore

- Created a proof of concept using advanced machine learning techniques to forecast company costs, improving accuracy over traditional time-series models.
- Reduced RMSE and MAE by approximately 20–25% while achieving higher R^2 scores compared to baseline regression models.
- Achieved a 20–30% improvement in prediction efficiency, supporting more reliable financial planning and budgeting decisions.

Fidelity Investments

May 2023 – Jul. 2023

Software Development Engineer Intern

Bangalore

- Built an internal dashboard using Angular, TypeScript, and Tailwind CSS to centralize status monitoring and key details of 30+ company applications.
- Integrated quick-access links and essential data views, streamlining access to critical information.
- Supported access to application information for more than five engineering teams.

Publications

Crowd Control and Monitoring using Deep Learning | *Computer Vision* — [IEEE Xplore Link](#)

Oct. 2024

- M S Janardhana Reddy, Amith C Kowshik, Chakshu Ramaswamy Sharma, Aayush, K J Bhanushree. Published in IEEE, Oct. 2024.
- Designed a deep-learning-based crowd monitoring system using YOLOv8-Pose for human pose detection, social distancing analysis, and crowd-density estimation.
- Integrated automated homography techniques to transform 2D pose detections into 3D coordinates, improving distance estimation and depth perception.
- Generated color-coded density maps and violation alerts to identify overcrowding and social-distancing violations in real-world environments.
- Achieved up to 98.7% accuracy on evaluated crowd-monitoring scenarios.
- This work won the **1st Prize** at the National Level Project Competition 2024 organized by the Department of Computer Science, Bangalore Institute of Technology.

Projects

Calibration Robustness under Label Noise in GBDTs | *ML Research*

Sep. 2026 – Present

- Designed three novel post-hoc calibration methods (NAC, RIC, IDC) for gradient boosted decision trees under label noise; benchmarked across 19 OpenML datasets, 3 GBDTs, 3 noise types, and 4 noise rates, yielding 35,235 experimental observations.
- Established RIC as lowest-ECE method on clean data (0.0125) and under severe asymmetric noise (0.0335); showed beta calibration degrades $8.1\times$, confirmed via Wilcoxon signed-rank tests with Holm correction (180/225 comparisons significant, $\alpha=0.05$).
- Extended multiclass calibration to $K > 2$ via full-matrix Dirichlet parameterisation and one-vs-rest isotonic decomposition with renormalisation.

Adaptive Grounding Diagnostics for Vision-Language Models | *Multimodal AI Research*

Sep. 2026 – Present

- Investigating whether lightweight classifiers trained on VLM internal activations can distinguish grounded, spuriously decodable, and ungrounded spatial reasoning regimes identified in arXiv:2606.31257.
- Designing an inference-time routing system for LLaVA-1.5-7B that applies regime-appropriate corrections without fine-tuning the underlying vision-language model.
- Evaluating cross-modal attention and hidden-state features across VSR, What'sUp, and CV-Bench using held-out regime classification, spatial accuracy, calibration, and accuracy-coverage analyses.

Ethical Dimensions of AI: A Critical Examination | *AI Policy & Ethics Research*

Aug. 2026 – Present

- Authoring an analytical research paper critically examining structural, political, and economic forces shaping a significant ethical issue in AI, including algorithmic bias, surveillance, data privacy, or governance of automated decision-making.
- Synthesizing scholarly literature and theoretical frameworks to analyze implications for equity, accountability, and public trust in AI systems.
- Connecting academic theory to concrete real-world cases to develop evidence-based recommendations for more just and responsible AI development and deployment.

On-The-Go Rental System | *Full-Stack*

Jan. 2023 – Feb. 2023

- Built a full-stack bike rental platform with REST APIs supporting user management, bike and terminal administration, rental workflows, and payment processing.
- Architected the system using four modular Django applications to separate business logic, improve maintainability, and support scalable feature development.
- Containerized the application with Docker Compose, enabling reproducible development and deployment environments.
- Implemented end-to-end payment and transaction management workflows, including rental payments, transaction history tracking, and administrative reconciliation.
- Created automated tests for core models and views to improve application reliability and streamline development.
- **Tech Stack:** Django, Python, MySQL, Docker, Docker Compose.

Extracurricular

Diminished 7th – Music Club, Bangalore Institute of Technology

Jun. 2022 – Jun. 2024

President & Band Lead

- Served as President and Band Lead of the college music club, overseeing club operations, organizing performances, and leading the band across college events and cultural fests.